

Preliminary Investigation of Spatial Variability of Sediment Fingerprinting Properties in Relation to Terrain Attributes at the Field-Scale

Maria Luna Miño¹, Alexander J Koiter¹, Taras E Lychuk¹, Arnie Waddel¹, Alan Moulin¹

^a*Brandon University, Masters in Environmental and Life Sciences, 270 18th St, Brandon, R7A 6A9*

^b*Brandon University, Department of Geography and Environment, 270 18th St, Brandon, R7A 6A9*

^c*Agriculture and Agri-Food Canada, Brandon Research and Development Centre, 2701 Grand Valley Road, Brandon, R7A 5Y3*

Abstract

Quantifying field-scale (40 ha) spatial variability in soil geochemical and colour (spectral reflectance) properties is important for sediment source fingerprinting. The main objectives of this study were to: 1) quantify the spatial variability and spatial dependence for a set of 15 properties (10 geochemical elements: Ca, Co, Cs, Fe, Li, La, Nb, Ni, Rb, Sr; and 5 colour coefficients: a^* , b^* , c^* , h^* , and x) measured from the <63 μ m fraction at an agricultural and forested site; and 2) assess how seven terrain attributes (elevation, SAGA wetness index, relative slope position, catchment area, vertical distance to channel network, plan curvature, profile curvature) relate to these properties through correlation and Random Forest (RF) regression. Across both sites, colour properties exhibited approximately normal distributions with low coefficients of variation (CV), whereas geochemical properties were more variable and skewed. Semivariograms indicated strong spatial dependence for most properties at the agricultural site (median range 210 m) and moderate dependence at the forested site (ranges 176–298 m). RF models performed better at the agricultural site; elevation, wetness and slope position were the top-ranked predictors for most properties. These results support DEM-based sampling strategies. In practice, sample spacing on the order of the semivariogram range (200–230 m) can suffice for mapping in simple agricultural settings, whereas stratified designs (e.g., near-stream vs. hillslope) may be preferable in more complex forested terrains.

Keywords: Soil geochemistry, Soil colour, Spatial analysis, Terrain attributes

*Corresponding author

0.1. Introduction

Spatial variation in soil properties arises from soil-forming factors (parent material, relief, biota, climate, time) and is further modified by land use and management practices. Characterizing the patterns and drivers of this variation is an important component of many agri-environmental studies. For example, to meet the desired level of precision for agronomic and environmental nutrient management plans the spatial variability in soil nutrients will influence the soil sampling design in terms of number and locations of soil samples (Kariuki et al., 2009; Starr et al., 1995). Similarly, accounting for spatial variability extends to sediment source fingerprinting, as unaccounted spatial heterogeneity within sources of sediment can compromise the source apportionment results (Du and Walling, 2017).

Sediment source fingerprinting is a watershed-scale technique that is used to quantify the relative proportions of sediment derived from unique sources. This technique uses naturally occurring biogeochemical properties as fingerprints (i.e., tracers) to link potential sources of sediment to downstream sediment. Investigating the spatial variability at a watershed-scale can be advantageous to identify, classify, and distinguish between potential sources of sediment (Pulley et al., 2017). However, investigating spatial variability at smaller scales is less common (e.g., Du and Walling, 2017; Pulley et al., 2018; Collins et al., 2019; Luna Miño et al., 2024) and remains a research priority (Collins et al., 2020). Within the sediment source fingerprinting approach field-scale ($<1 \text{ km}^2$) variability is important for (i) designing efficient, representative source sampling, (ii) integrating spatial variability with erosion/delivery gradients for mixing models, and (iii) selecting fingerprints supported by process-based reasoning. Previous work has highlighted spatial variability issues and the need for more robust, process-informed fingerprint selection and sampling strategies.

Characterization of fingerprint properties requires representative sampling, particularly because variability in some properties is not random but follows systematic spatial patterns across the landscape. For example, the pattern of fallout radionuclides will reflect the long-term patterns of soil erosion and deposition (Wilkinson et al., 2015). Designing and implementing source sampling campaigns must consider this, as the sampling design used has been shown to influence the characterization of a wide range of commonly used fingerprints (Luna Miño et al., 2024). The lack of standardization in sediment source sampling designs makes it difficult to compare results across studies and creates uncertainties in the interpretation of results (Luna Miño et al., 2024).

Characterization of sources is further complicated by variations in erosion and sediment delivery rates across the landscape. Many mixing models use well defined inputs (sources) and outputs (sediment) characterized by mean and standard deviation, but do not consider the spatial distribution of fingerprint properties. This is a limitation of the methodology as samples collected closer

to and more hydrologically connected with the stream network may better represent the source, even if they deviate from the mean value. Strategic/judgment sampling targeting the likely to eroded areas may improve representativeness, but it prevents an overview of the broader spatial patterns, limiting the interpretation of landscape processes across the entire source area. There has been progress using information on erosion rates to calculate an erosion rate-weighted mean (Wilkinson et al., 2015; Du and Walling, 2017) and using spatially interpolated maps of fingerprint values to provide a finer resolution of the fingerprint variability within each source (Haddadchi et al., 2019). Characterizing the geomorphic, hydrologic, and biochemical processes underlying spatial variability supports the selection of reliable fingerprints and informs sampling design for source characterization. While many studies use a statistical-based approach (Collins et al., 1997) to select suitable fingerprints there are concerns that this approach may result in the inclusion of false positives (i.e., type I error) in which a fingerprint is incorrectly identified as having discriminatory power between sources when it does not, or non-conservative fingerprints (Koiter et al., 2013a). Consequently, there has been a call for the inclusion of a process-based (e.g., weathering, erosion) or geologic-based explanation of the fingerprints selected to address these concerns (Collins et al., 2020). This remains challenging due to limited data on soil properties, particularly with geochemical properties, as routine lab analysis often measures more than 50 elements. The spatial patterns of some soil properties are well studied because of their agronomic importance or ability to infer other important soil properties and processes and include fallout radionuclides (e.g., ^{137}Cs , ^7Be ; Ritchie et al. 1970), plant nutrients (e.g., N, P; Vasu et al. 2017), soil colour (e.g., hue, value; Viscarra Rossel et al. 2006), major non-acid forming cations (e.g., Ca, Na; Sun et al. 2021a). In contrast, the processes that control the distribution of rare earth elements and trace metals, are less well studied or tend to be site-specific, making it difficult to draw generalizations.

Terrain attributes such as elevation, slope curvature, slope position, and soil wetness indices have been shown to be useful information in the characterization and modelling of a range of soil properties including soil moisture (Beaudette et al., 2013), texture (Kokulan et al., 2018), colour (Brown et al., 2004), organic matter (Zhang et al., 2012), conductivity (Umali et al., 2012), and geochemistry (Lima et al., 2023). Similar techniques may provide additional insight into the pedologic and geomorphic processes that drive the observed patterns of fingerprint properties within a given source. Digital elevation models (DEMs) are more widely available through public databases and can be generated using drone imagery, whereas soil property data are more limited in availability and spatial coverage. As a result, terrain attributes derived from DEMs offer a practical and more readily available basis for guiding efficient and spatially informed soil sampling designs. This study builds on the previous work of Luna Miño (2024) where the impact of three different sampling designs on the characterization of source materials, within the framework of the sediment fingerprinting approach, was assessed. This study expands that study by using the data from

grid sampling approach to assess the spatial autocorrelation, and identify important terrain attributes driving the observed patterns. The objectives of this study were to: (1) quantify the spatial variability and spatial dependence of 15 fingerprint-relevant soil properties across an agricultural and forested field site; and (2) test the relation between these soil properties and seven terrain attributes (derived from DEMs), using correlation and Random Forest regression to identify terrain controls and their relative importance. Together, these objectives address how terrain attributes may be used to understand spatial distributions of soil properties and help guide sampling design.

0.2. Methods

0.2.1. Site description

Two sites within ~4 km of each other in the Wilson Creek Watershed (WCW), near McCreary, Manitoba, Canada were sampled (Figure ??). The mixedwood forest site (50°43'35"N, 99°33'36"W; 369.2 masl) in Riding Mountain National Park consists of white spruce (*Picea glauca*), black spruce (*Picea mariana*), balsam fir (*Abies balsamea*), larch (*Larix laricina*), and trembling aspen (*Populus tremuloides*), with soils likely part of the Grey Wooded soil association (Luvisol) developed on boulder till of mostly shale with some limestone and granitic rocks (Ehrlich et al., 1958). This site has a local relief of 17.9 m and moderately complex terrain, dissected by Wilson Creek, a meandering stream with a gradient of approximately 0.006 m m⁻¹, featuring a small floodplain 5 to 25 m wide and adjacent upland areas (Figure ??). The agricultural site (50°45'15"N, 99°30'49"W; 309.9 masl) includes grain and forage rotations and is mapped to the Edwards Soil Series (Cumulic Regosol) with silty clay loam to silty clay soils developed on alluvial deposits (Ehrlich et al., 1958). The site has relatively simple terrain with a local relief of 2.8 m and drains toward the northeast corner of the field, where Wilson Creek runs along the northern edge as a straight, engineered channel (Figure ??). The region has a Dfb climate (Beck et al., 2018), with a mean annual temperature of 3.0°C, annual precipitation of 539 mm (~27% as snow) (Environment and Canada, 2024), and snowmelt-dominated runoff, with about 80% occurring in May and June (MacKay, 1970).

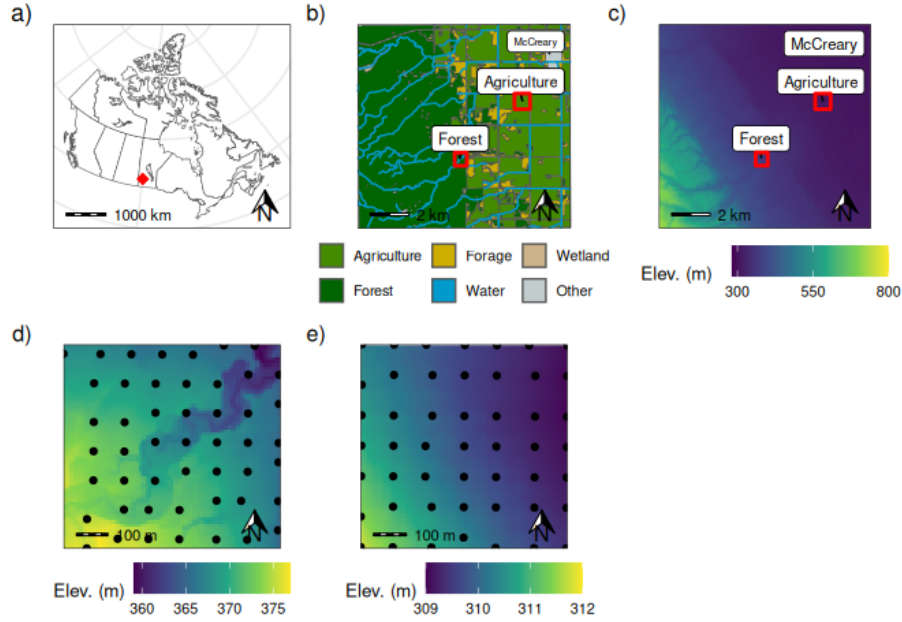


Figure 1: a) Map showing the location of the study sites within Canada. Location of the two study sites and nearby town of McCreary, and regional b) land use, and c) topography. Local topography and soil sampling locations at the d) forested site and e) agricultural site

Source: [Research Site Locations](#)

0.2.2. Soil sampling and analysis

At each site, 49 surface samples were collected on a 7×7 grid (100 m spacing) (Figure ??). Forested samples were collected 0–5 cm below the LFH to characterize the mineral A horizon, minimizing organic layer heterogeneity; agricultural samples were 0–15 cm to reflect mixing by tillage and operations and to capture the actively managed surface layer. Samples were dried, homogenized, and sieved to $<63 \mu\text{m}$ to minimize grain-size and organic matter effects across sites and to align with established fingerprinting protocols (Lacey et al., 2017). Samples were analyzed for a broad suite of geochemical elements using aqua-regia digestion with ICP-MS (ALS Mineral Division, North Vancouver, BC, Canada). Spectral measurements were made with a spectroradiometer (ASD FieldSpecPro Malvern Panalytical Inc Westborough MA 01581, United States). Spectral reflectance measurements were taken in 1 nm increments over the 0.4–2.5 μm wavelength range. Both samples and the Spectralon standard (white reference) were illuminated with a white light source using a halogen-based lamp (12 VDC, 20 Watt). Dark-current correction and white reference were completed between samples. Fifteen colour coefficients (Table S??) were computed from each sample, where each sample represents the average of 10

spectra, following published methods. (Koiter, 2021; Boudreault et al. (2018). Based on the results of Luna Miño (2024), a composite fingerprint consisting of 10 geochemical elements (Ca, Co, Cs, Fe, Li, La, Nb, Ni, Rb, and Sr) and five colour coefficients (a^* , b^* , c^* , h^* , and x) were identified as providing a strong discrimination between the agricultural and forested surface soils. These fifteen soil properties are the focus of the detailed spatial analysis detailed in this study.

0.2.3. Geospatial and terrain analysis

All geostatistics were performed with ArcGIS Pro (v 3.3.0 Esri, 2024). Semi-variograms were used to quantify spatial correlation for each of the 15 soil properties. The optimization tool, based on minimizing the mean square error, was used to parameterize the semivariogram model. In all cases the Stable model type was used as it offers greater flexibility in fitting the spatial correlation structure. The semivariograms were visually inspected to ensure the automated fits were reasonable. Kriging was used to interpolate and generate maps of each soil property. The exploratory interpolation tool (Geostatistical Analyst extension) was used to select the kriging type with the highest ranked prediction accuracy. Cross-validation was used to assess the performance of the spatial interpolation and the measured and predicted values were plotted and the r^2 was reported.

The forested digital elevation model (DEM) was derived from Natural Resources Canada’s High-Resolution DEM Mosaic (HRDEM) (Natural Resources Canada, 2024), while the agricultural DEM was generated using UAV photogrammetry processed in Agisoft Metashape (v1.8.2 Agisoft, 2021) with ground control points. Both DEMs were interpolated to 1 m resolution grids using ordinary kriging. Terrain attributes including elevation, plan and profile curvature, SAGA wetness index, catchment area, relative slope position, and vertical distance to channel network were computed in SAGA GIS (SAGA v2.1.4 Conrad et al., 2015) (Table S??). The native resolution between the HRDEM and UAV-derived DEM differ, which may affect curvature metrics. All rasters were resampled to 10 m resolution (terra v1.8.29 Hijmans, 2024) prior to statistical analysis to harmonize spatial support.

0.2.4. Data analysis

All subsequent statistical analysis was undertaken using R statistical Software v4.4.0 (R Core Team, 2024) through RStudio Integrated Development Environment v2024.04.2 (RStudio, 2024). Plots and maps were created using the R package ggplot2 v 3.5.2 (Wickham, 2016). Univariate summaries included CV classes (low <15%, moderate 15–35%, high 35–75%, very high >75%) and skewness categories (symmetric: -0.5 to 0.5 ; moderate: -1.0 to -0.5 or 0.5 to 1.0 ; high: <-1.0 or >1.0).

Random forest regression (RF) (randomForest v4.7.1.2 Liaw and Wiener, 2002) was used to assess the relative importance of the terrain attributes on the spatial distribution of soil properties. The dataset was randomly split into train-

ing (70%) and testing (30%) datasets. Multicollinearity among the terrain attributed was assessed using the Variance Inflation Factor with a threshold of eight (usdm v2.1.7 Naimi et al., 2014). The number of variables randomly sampled as candidates (mtry) at each split within the RF model was tuned using the training data set (caret v7.0.1 Kuhn and Max, 2008). The number of trees to grow was set to 500 to ensure the error has stabilized and that additional trees provide diminishing returns. Model performance was assessed using the Mean Square Error (MSE) and percent variance explained for both the training (Out of Bag Error) and the validation data sets. To test the model, measured and predicted values were plotted and the r^2 and MSE were calculated using the testing data set. Variable importance is reported as percent increase in MSE.

0.3. Results

0.3.1. Univariate summary

The univariate analysis demonstrated some variability patterns between colour and geochemistry properties and sites (Table ??). All five colour coefficients display a symmetric distribution and low CV, with the exception of a moderate CV for b^* in the forested site. Geochemistry data display greater variability. At the agriculture site, eight elements showed a symmetric distribution and two displayed moderate skew (Li, Co); six had low CV, two had moderate (Cs, Rb), and two had high (Ca, Sr). The forest site had six elements displaying symmetric distributions, three with moderate skew (Fe, Nb, Sr), and one with a large skew (Ca); three had low CV, five had moderate, one had high (Sr) and one very high (Ca). Overall this highlights that the measured geochemical properties more skewed and variable compared to the colour properties across both sites.

The mean plan and profile curvature measurements for both sites are near zero, indicating an area of sediment transit and not accumulation or erosion (Table ??). The agricultural site had a higher SAGA Wetness Index but the forested site had a larger range in values and exhibited greater variability. The forested site exhibited a smaller mean Relative Slope Position value (streams and depressional areas) and a smaller Vertical Distance to Channel Network, and for both terrain attributes a greater variability as compared to the agricultural reflecting the presence of the stream crossing the forested site.

Table 1: Summary univariate statistics of selected geochemical and colour soil properties for each site ($n = 49$). Geochemical concentrations are reported in ppm, except Ca and Fe(%).

Property	Mean	SD	Max	Min	Skewness	CV
Agriculture						
Ca	4.00	2.19	8.78	0.95	0.28	54.66
Co	8.76	0.83	10.60	7.50	0.52	9.48
Cs	0.75	0.15	1.07	0.47	0.18	19.93
Fe	1.92	0.09	2.11	1.71	-0.25	4.70
Li	15.62	1.42	19.80	12.80	0.62	9.11

Table 1: Summary univariate statistics of selected geochemical and colour soil properties for each site (n = 49). Geochemical concentrations are reported in ppm, except Ca and Fe(%).

Property	Mean	SD	Max	Min	Skewness	CV
La	18.23	1.22	20.20	15.50	−0.29	6.71
Nb	0.59	0.06	0.73	0.46	0.45	9.67
Ni	29.63	2.72	35.70	25.00	0.36	9.17
Rb	18.43	4.33	26.70	10.20	0.24	23.48
Sr	91.31	38.98	163.50	38.60	0.09	42.69
a*	3.38	0.32	4.15	2.59	−0.03	9.53
b*	8.84	0.97	10.59	6.69	−0.18	11.00
c*	9.47	1.02	11.32	7.17	−0.19	10.74
h*	1.20	0.01	1.23	1.18	0.19	1.12
x	0.47	0.00	0.48	0.47	0.06	0.46
Forest						
Ca	1.89	1.53	5.46	0.47	1.07	81.12
Co	6.76	1.39	9.60	4.00	0.03	20.62
Cs	0.55	0.12	0.78	0.34	0.25	21.73
Fe	1.18	0.13	1.46	0.83	−0.58	11.24
Li	6.47	0.90	8.60	4.30	−0.02	13.89
La	15.00	2.60	21.80	10.30	0.33	17.31
Nb	0.37	0.06	0.56	0.17	−0.68	17.10
Ni	18.09	3.90	28.00	11.00	0.33	21.55
Rb	13.83	1.85	18.10	9.90	0.27	13.40
Sr	32.43	12.60	64.20	15.30	0.98	38.87
a*	5.73	0.41	6.56	4.41	−0.38	7.10
b*	12.47	2.01	15.91	8.02	0.22	16.11
c*	13.74	1.94	17.00	9.15	0.15	14.15
h*	1.13	0.05	1.23	1.06	0.34	4.13
x	0.49	0.00	0.49	0.48	−0.21	0.47

Source: [Univariate summary](#)

Table 2: Summary univariate statistics of terrain attributes for each site (n = 49).

name	mean	sd	max	min	skewness	cv
Agriculture						
Catchment Area	454.54	1,033.10	6,442.41	8.41	4.46	227.29
Vert. Dist. Channel	0.06	0.05	0.22	0.01	1.14	81.39
Elevation	309.96	0.69	311.78	309.03	0.68	0.22
Plan Curvature	0.00	0.00	0.00	0.00	−0.10	−3,103.61
Profile Curvature	0.00	0.00	0.00	0.00	−0.31	−1,309.29
Rel. Slope Position	0.67	0.33	1.00	0.05	−0.73	49.05
SAGA Wetness Index	9.57	0.81	10.88	7.86	−0.23	8.50

Table 2: Summary univariate statistics of terrain attributes for each site (n = 49).

name	mean	sd	max	min	skewness	cv
Forest						
Catchment Area	987.43	3,567.87	23,581.83	4.34	5.53	361.33
Vert. Dist. Channel	0.48	0.49	2.83	0.05	2.68	103.92
Elevation	368.81	3.51	375.93	360.86	0.09	0.95
Plan Curvature	0.00	0.01	0.03	-0.01	2.25	396.75
Profile Curvature	0.00	0.02	0.05	-0.07	-1.66	2,724.16
Rel. Slope Position	0.23	0.23	0.87	0.02	1.39	100.24
SAGA Wetness Index	5.79	1.17	8.00	2.64	-0.56	20.17

Source: [Univariate summary](#)

0.3.2. Spatial analysis

Within the agricultural site all 15 properties displayed spatial autocorrelation; many with strong dependence (Table ??). Semivariogram ranges spanned ~185–580 m (median 210 m). Ca and Rb exhibited large ranges (580 m and 551 m), while colour metrics such as b^* and c^* had ranges ~199 m. The nugget (Co) was small for all soil properties (<1.5). A few of the soil properties presented a pattern that roughly matches (e.g., Rb) or mirrors (e.g., Ca, Sr) the overall topography of the site with a gradation between the highest point in the south-west corner towards the lowest points in the north-east (Figure ??). Other properties appear to have more localized high and low concentrations/values (e.g., c , h). *Within the forested site nine properties showed spatial dependence (moderate to strong) with ranges 176–298 m; Fe, Rb and colour a , b , c , x lacked detectable autocorrelation, consistent with more complex depositional/topographic features found at this site (Table ??). This complexity also resulted in poorer spatial interpolation performance compared to the agricultural site. The nugget (Co) was generally small for most soil properties (<2) with the exception of La and Ni. Overall, the influence of the channel and floodplain environment can be seen in the pattern of the nine soil properties (Figure ??).*

Table 3: Geostatistical parameters of the fitted semivariogram models of selected colour and geochemical properties within the agricultural and forested sites.

Property	Kriging Type*	Nugget (Co)	Sill (Co + C)	C/(C + Co) (%)	Range (m)	$r^{2\dagger}$	Spatial Cla
Agriculture							
Ca	Universal	0.0	7.2	100	580	0.9	
Co	Simple	0.0	0.7	100	208	0.4	
Cs	Ordinary	0.0	0.0	100	210	0.5	
Fe	Ordinary	0.0	0.0	100	185	0.2	
Li	Universal	0.3	1.5	81	185	0.6	
La	Simple	0.4	1.0	56	308	0.5	

Table 3: Geostatistical parameters of the fitted semivariogram models of selected colour and geochemical properties within the agricultural and forested sites.

Property	Kriging Type [*]	Nugget (Co)	Sill (Co + C)	C/(C + Co) (%)	Range (m)	r ^{2†}	Spatial Cla
Nb	Universal	0.0	0.0	91	210	0.7	
Ni	Ordinary	1.4	8.9	84	352	0.6	
Rb	Ordinary	1.4	27.6	95	551	0.9	
Sr	Ordinary	0.9	900.2	100	220	1.0	
*a**	Ordinary	0.4	1.0	59	288	0.3	
*b**	Simple	0.2	0.9	83	199	0.3	
*c**	Simple	0.1	0.9	87	199	0.3	
*h**	Simple	0.0	1.1	100	185	0.2	
<i>x</i>	Simple	0.4	1.0	58	220	0.1	
					Forest		
Ca	Ordinary	1.6	2.7	41	269	0.2	
Co	Ordinary	0.0	2.1	100	298	0.1	
Cs	Ordinary	0.0	0.0	83	237	0.2	
Li	Ordinary	0.0	0.8	100	222	0.3	
La	Ordinary	3.1	7.4	59	176	0.1	
Nb	Ordinary	0.0	0.0	51	224	0.2	
Ni	Universal	6.7	15.8	57	187	0.2	
Sr	Simple	0.4	1.0	65	229	0.4	
*h**	Universal	0.0	0.0	100	230	0.3	

^{*} Stable model type was used for all analysis. Models are all i

[†] Cross-validation of the measured and predicted value

[‡] Strong spatial dependency (C/(C + Co) % >75); Moderate spatial dependency (C/(C + Co) % between 75

Source: [Semivariograms](#)

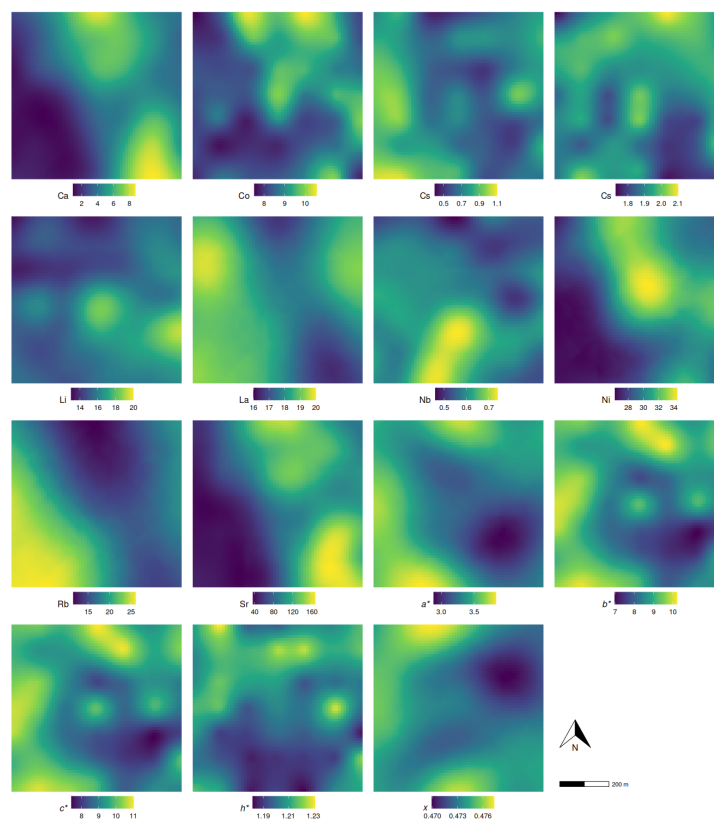


Figure 2: Kriged maps of select colour and geochemical properties and elevation across the agricultural site.

Source: [Soil property mapping](#)

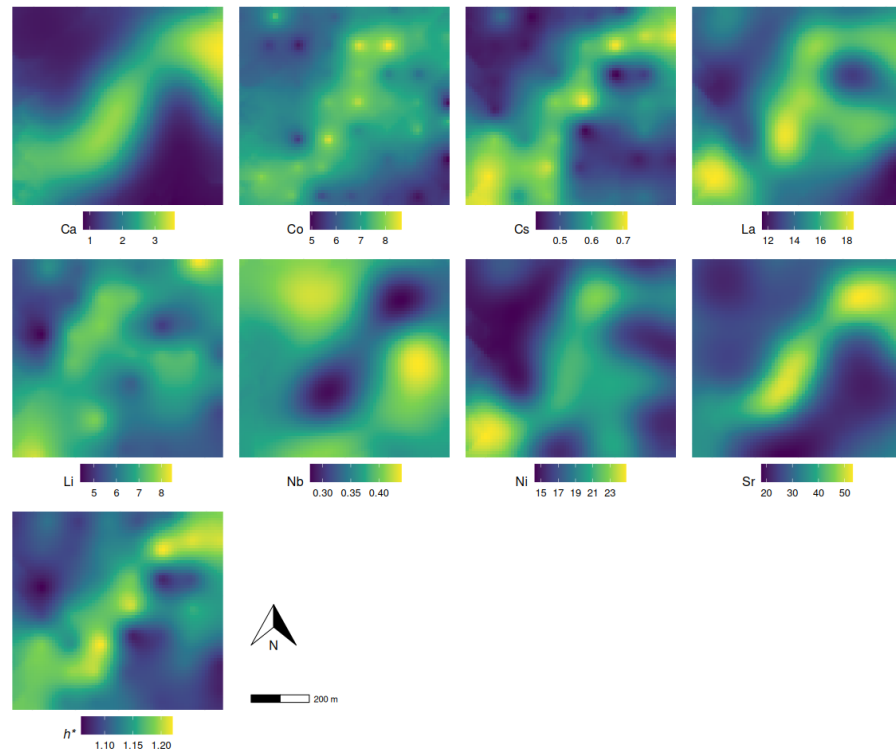


Figure 3: Kriged map of select colour and geochemical properties and elevation across the forested site.

Source: [Soil property mapping](#)

With the exception of h^* all the soil properties measured at the agricultural

site were significantly ($p < 0.05$) correlated with the elevation and the wetness index (Table ??). A few soil properties (Li, Ni, Sr, a^* and x^*) were correlated relative slope position and vertical distance to the channel, and there was little to no correlation with catchment area and curvature attributes. Overall there was little correlation between soil properties and terrain attributes at the forested site (Table ??).

Table 4: Pearson's correlation coefficients for select soil properties and terrain attributes (n=49).

Property	Elevation	SAGA Wetness Index	Rel. Slope Position	Vert. Dist. Channel	Catchment Area
Agriculture					
Ca	-0.71***	0.61***	NS	-0.32*	NS
Co	-0.52***	0.56***	NS	NS	NS
Cs	0.56***	-0.5***	NS	NS	NS
Fe	NS	0.33*	NS	NS	NS
Li	-0.39**	0.38**	-0.56***	-0.39**	NS
La	0.51***	-0.39**	NS	NS	NS
Nb	0.41**	-0.41**	NS	NS	NS
Ni	-0.72***	0.73***	-0.29*	-0.32*	NS
Rb	0.78***	-0.72***	NS	0.32*	NS
Sr	-0.78***	0.7***	-0.39**	-0.38**	NS
*a**	0.6***	-0.49***	0.43**	0.36*	NS
*b**	0.39**	-0.31*	NS	NS	NS
*c**	0.4**	-0.31*	NS	NS	NS
*h**	NS	NS	NS	NS	NS
x	0.48***	-0.55***	0.5***	0.41**	-0.37**
Forest					
Ca	NS	-0.58***	NS	0.45**	0.35*
Co	NS	NS	NS	NS	NS
Cs	NS	NS	NS	NS	0.33*
Li	NS	NS	NS	NS	NS
La	NS	NS	NS	NS	NS
Nb	NS	0.57***	NS	-0.45**	NS
Ni	NS	NS	NS	NS	NS
Sr	-0.35*	-0.58***	NS	0.44**	0.31*
*h**	NS	NS	NS	0.31*	0.33*

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; NS = non-significant at $p = 0.05$

Source: [Univariate summary](#)

No multicollinearity among the terrain attributes at either site was detected and all seven attributes were retained for all analysis. The mtry parameter selected for each model varied with most using a value of four. The RF regression model showed variable explanatory power and performance across soil proper-